

ITAI 1378 - MIDTERM PROJECT PROPOSAL

Due: October 30 (Week 10) - Submit before class

Worth: 25% of your final grade

OBJECTIVE

Create a clear, realistic plan for your computer vision project. Show us you understand:

- What problem you're solving
- How you'll solve it
- That your plan is achievable with available resources

Remember: You're planning to BUILD something, not write about it!

WHAT TO SUBMIT

1. Presentation Slides (PDF or PPTX)

- 8-10 slides
- You'll present for 5 minutes in class

2. GitHub Repository

- Basic repository structure
- README.md with your proposal details
- Initial exploratory code (optional but recommended)

That's it! No long written reports needed.

PRESENTATION STRUCTURE (8-10 Slides)

Slide 1: Title

- Project name
- Team member names
- Which tier: Tier 1, 2, or 3?

Slide 2: The Problem

Answer these questions:

- What real-world problem are you solving?
- Who cares about this problem?
- Why is it important?

Example: Problem: Warehouse workers spend 2 hours/day manually counting inventory.

Who Cares: Warehouse managers, logistics companies

Why Important: Costs \$50,000/year in labor, causes counting errors

Slide 3: Your Solution (Simple Overview)

- What will your system do?
- How will it solve the problem?
- One sentence description

Example: Solution: Automated inventory counting system using object detection

How: Camera takes shelf photos → YOLO detects products → Counts items → Generates report

Slide 4: Technical Approach

Specify:

- **CV Technique:** Object detection? Classification? Segmentation? VLM?
- **Model:** YOLOv8? ResNet? CLIP? LLaVA?
- **Framework:** PyTorch? TensorFlow? Hugging Face?
- **Why this approach?** (1-2 sentences)

Slide 5: Data Plan

- **Source:** Where will you get images/videos?
 - Public dataset? (COCO, ImageNet, Roboflow)
 - Collect your own? (Phone camera? Web scraping?)
 - Synthetic data?
- **Size:** How many images? (100? 1000? 10,000?)
- **Labels:** Do you need to label it? How?

Slide 6: System Diagram

[Input] → [Processing Step 1] → [Processing Step 2] → [Output]

Slide 7: Success Metrics

How will you measure success?

- **Primary Metric:** Accuracy? Precision? Recall? Speed?
- **Target:** "90%+ detection accuracy"
- **Secondary Metrics:** Processing time, FPS, etc.

Example: Success = 90%+ detection accuracy AND <1 second per image

Slide 8: Week-by-Week Plan

Week	Tasks	Milestone
10 (Oct 30)	Get dataset, set up environment	Dataset ready
11 (Nov 6)	Train/fine-tune model	Model working
12 (Nov 13)	Test and improve	Good accuracy
13 (Nov 20)	Create demo, make video	Demo ready
14 (Nov 27)	Final testing, documentation	Everything done
15 (Dec 4)	Present!	🎉

Slide 9: Challenges & Backup Plans

What could go wrong? What's your Plan B?

Example:

Challenge: What if I can't collect enough data?

Plan B: Use public dataset from Roboflow instead

Challenge: What if accuracy is too low?

Plan B: Try data augmentation or larger model

Slide 10: Resources Needed

- **Compute:** Colab? Heidi? Kaggle?
- **Tools:** PyTorch? Ultralytics? Hugging Face?
- **Cost:** 0?5? \$10?

GITHUB REPOSITORY SETUP

Required Structure:

```
project-name/
├── README.md           ← MOST IMPORTANT!
├── requirements.txt    ← Python packages
└── notebooks/
```

```

├── notebooks/
│   └── 01_exploration.ipynb ← Initial experiments (optional)
├── data/
│   └── README.md ← Dataset info
└── docs/
    └── proposal.pdf ← Your slides

```

README.md Template:

[Your Project Name]

Team Members (if applicable)

- [Name 1] - [Email]
- [Name 2] - [Email] (if applicable)
- [Name 3] - [Email] (if applicable)

Project Tier

Tier [1/2/3]: [Brief justification why this tier]

Problem Statement

[2-3 sentences describing the problem you're solving]

Solution Overview

[2-3 sentences describing your solution]

Technical Approach

- **CV Technique:** [Object Detection / Classification / Segmentation / VLM / etc.]
- **Model:** [YOLOv8 / ResNet50 / CLIP / etc.]
- **Framework:** [PyTorch / TensorFlow / Hugging Face]
- **Why this approach:** [1-2 sentences]

Dataset

- **Source:** [COCO / Custom / Roboflow / etc.]
- **Size:** [Number of images/videos]
- **Labels:** [What you're detecting/classifying]
- **Link:** [URL if public dataset]

Success Metrics

- **Primary Metric:** [Accuracy / mAP / F1-score / etc.]
- **Target:** [e.g., "90% accuracy"]
- **Secondary Metrics:** [Speed, etc.]

Week-by-Week Plan

- **Week 10:** [Tasks]
- **Week 11:** [Tasks]
- **Week 12:** [Tasks]
- **Week 13:** [Tasks]
- **Week 14:** [Tasks]
- **Week 15:** Presentation

Resources Needed

- **Compute:** [Colab / Heidi / Kaggle]
- **Cost:** [0 / 5 / \$10]
- **APIs:** [If using paid APIs]

Risks & Mitigation

Risk	Probability	Mitigation
[Risk 1]	High/Med/Low	[Plan B]
[Risk 2]	High/Med/Low	[Plan B]

AI Usage Log

[Will be updated as we use AI tools]

Current Status

- ☒ Repository created
- ☒ Proposal written
- ☐ Dataset acquired
- ☐ Model training started
- ☐ Demo created
- ☐ Final presentation ready